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Chapter 10

Planetary Astrophysics of Small Bodies

David Jewitt

*Department of Earth, Planetary and Space Sciences,
University of California, Los Angeles, USA*

jewitt@ucla.edu

This is an intentionally brief overview of Solar System science as seen from the perspective of an observational planetary astronomer. Instead of following the format used in my lectures for the school, I thought it would be more fun to focus on a set of topics that illustrate current problems and areas of activity in planetary science.

1. Introduction

The small bodies of the Solar System are carriers of information from the earliest epochs and, therefore, objects of great scientific interest. Observationally, the small body populations tend to be difficult to study, both because small means “faint”, and because many of the bodies of greatest interest (the Kuiper belt objects, the Centaurs, the Trojans) are far-away residents of the middle and outer Solar System, rendering them fainter still. To add insult to injury, the albedos of

many of the most primordial objects (notably the nuclei of comets but also the Trojans and many Kuiper belt objects), tend to be very small (few percent), making them even fainter still. This is why we can study self-luminous objects at the edge of the universe, but we can barely glimpse what's in the Kuiper belt only 100 AU away. It's also why the study of the small body populations is very fresh and new.

2. The Comets

Comets are dynamically divided into two main groups, the short-period comets (SPCs) and the long-period comets (LPCs). The easy-to-remember but rather arbitrary distinction between them is that the SPCs have periods $P < 200$ years while the LPCs have $P > 200$ years. When separated by this simple criterion, the LPCs and the SPCs are remarkably dynamically distinct.^a The SPCs occupy a disk-like distribution with modest inclinations and eccentricities and share the prograde motion of the planets around the Sun. The LPCs occupy an isotropic distribution, with equal numbers of prograde and retrograde orbits, and equal numbers arriving from the northern and southern hemispheres, as far as can be told. They have large eccentricities, up to $e = 1$. Objects in a third group, the Halley-type comets, (HTCs) are dynamically intermediate. Most are prograde but some retrograde HTCs exist. A famous example is the class prototype, 1P/Halley, which has $P = 76$ years like the SPCs but $i = 162^\circ$ like the LPCs.

Seemingly less arbitrary but more involved classification systems exist, notably one based on the Tisserand parameter, T_J . This is a constant of the motion in the restricted, circular three-body (Sun–Jupiter–comet) problem, defined by

$$T_J = \frac{a_J}{a} + 2 \left[(1 - e^2) \frac{a}{a_J} \right]^{1/2} \cos(i), \quad (1)$$

^aThe selection of 200 years is also not particularly critical. $P = 100$ and 1,000 years give roughly the same separation.

where a , e and i are, respectively, the semi-major axis, eccentricity and inclination of the comet orbit and $a_J = 5.2$ AU is the semi-major axis of Jupiter's orbit. By this parameter, the LPCs have $T_J \leq 2$ while the SPCs have $2 < T_J \leq 3$. Although it looks superficially more formal, the T_J classification is itself not particularly rigorous because the solar system is not accurately represented by the restricted, circular three-body approximation (e.g., Jupiter's orbit is not circular, there are many planets other than Jupiter, they have non-zero inclinations, and so on). Happily, the period-defined and T_J -defined comet groups are roughly consistent.

Most comets arrive in the inner Solar System from two distinct storage locations, known as the Kuiper belt (KB) and the Oort cloud (OC) (e.g., Ref. 1). The KB is the source of most SPCs (specifically the Jupiter family comets). It has an inner edge carved by Neptune at ~ 30 AU and extends for thousands of AU outwards. It consists of four dynamically distinct components (the classical, resonant, scattered and detached Kuiper belt objects), each with a story to tell about the origin and dynamical evolution of the Solar System. The KB contains roughly 10^5 objects larger than 100 km and perhaps 10^{10} larger than 1 km, but the exact population is not well-defined for two reasons. First, KBOs smaller than a few $\times 10$ km are too faint to be reliably detected in existing ground-based surveys. Second, the flux density, f [$\text{W m}^{-2} \text{Hz}^{-1}$], of an object in scattered sunlight suffers from the inverse-square law twice (once from Sun to the object and again from the object to Earth), and so varies with distance from the Sun, r_H , as $f \propto r_H^{-4}$. A given object at 50 AU appears 10^4 times fainter than it would at 5 AU (Jupiter's orbit). In practice, this steep dependence means that we can say little about the contents of the KB beyond $r_H = 50$ –100 AU and there is room for almost any non-luminous object beyond a few hundred AU. Most SPCs originate in the scattered disk component of the KB. They are ejected as a result of dynamical chaos, at a rate estimated at a few per century.

The OC is the source of LPCs. It holds $\sim 10^{12}$ comets in a spherical swarm that is not sharp-edged but which has a characteristic diameter $\sim 10^5$ AU.¹ This is a large fraction of the distance to the nearest stars ($\sim 2 \times 10^5$ AU). Indeed, the extent of the cloud is set by erosion from the gravity of passing stars coupled with tides caused by the lop-sided distribution of mass in the Milky Way galaxy. The properties of the OC are inferred from the orbits of LPCs that have been

deflected out of the cloud and enter the planetary region. A passing star of mass M_* , relative velocity V_* and with impact parameter (minimum distance) d perturbs the velocity of an orbiting OC comet by

$$\Delta V \sim \left(\frac{2GM_*}{V_* d} \right)^{1/2}. \quad (2)$$

Substituting $G = 6.6 \times 10^{-11} \text{ N kg}^{-2} \text{ m}^2$, $M_* = M_\odot = 2 \times 10^{30} \text{ kg}$, $V_* = 20 \text{ km s}^{-1}$ and $d = 1 \text{ pc}$ ($3 \times 10^{16} \text{ m}$) gives $\Delta V \sim 1 \text{ m s}^{-1}$. For comparison, the Keplerian velocity at 10^5 AU from the Sun is $V_K \sim 100 \text{ m s}^{-1}$. Since field stars pass the Sun from random directions, the ΔV “kicks” accumulate randomly, and the total velocity change is equal to V_K after about $N \sim (V_K/(\Delta V))^2$ stellar flybys. Evidently, $N \sim 10^4$, which takes about a billion years (the process is aided by the galactic tide and the degree of orbit randomization depends as well on the comet semi-major axis and local stellar density²). On this and longer timescales, the OC, which began as a flattened disk sharing the midplane of the planets, was isotropized into a spherical cloud. Consequently, considerable uncertainties remain concerning the radial structure, the degree of isotropy, the mass and other parameters, and we do not know how or where the KB and the OC meet, except by theoretical conjecture.

A third but much smaller comet reservoir has been recently located in the asteroid belt,³ where ice-rich asteroids unexpectedly survive. We will discuss these “main-belt comets” in another section. The three reservoirs are illustrated schematically in Fig. 1. The end-fates of comets are to impact planets or the Sun, to be ejected from the Solar System, or to spontaneously disintegrate. In between their demise and their origin in the source reservoirs, they are classified (for largely historical reasons) into numerous different groups. Damocloids are defunct LPCs.⁴ Centaurs are recently escaped KBOs in the process of crossing the giant planet region of the Solar System. Centaurs surviving the jump across Jupiter’s orbit are relabeled Jupiter family comets (JFCs). Defunct (i.e., thermally evolved) JFCs are known as dJFCs or Asteroids on cometary orbits (ACOs). The names don’t matter much; what is important is that the comets flow into the planetary region from distinct source reservoirs that have had different dynamical and thermal histories.

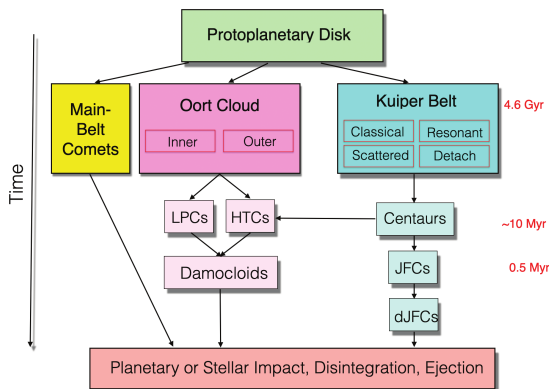


Fig. 1. Schematic diagram showing the relationships between various cometary populations discussed in the text. The timescales on the right indicate the approximate lifetimes in each stage. Acronyms are defined in the text.

2.1. Destruction of the comets

While we have made real progress in identifying and characterizing the different source regions of the comets, the ways in which they are destroyed are still not clear. As noted above, the likely processes are impact destruction, dynamical ejection to the interstellar medium (never to be seen again) and physical disintegration. Numerical integrations show that scattering by the planets limits the Jupiter family comets to a dynamical half life $\tau_d \sim 0.4$ Myr. On this timescale, the comets either strike a planet or the Sun or are ejected to infinity. The simulations also show that successive scattering interactions tend to pump up the inclinations of the orbits relative to the inclination distribution of objects arriving from the Kuiper belt. In order to prevent the model inclination distribution from growing fatter than the observed inclination distribution, the evolution must be truncated after about $\tau \sim 12,000$ years, which gives a measure of the “physical lifetime” of the comets. The question is “why is $\tau \ll \tau_d$?” or, equivalently, “what determines τ ?”.

Ice exposed to sunlight sublimates into the interplanetary vacuum, at a rate f_s ($\text{kg m}^{-2} \text{s}^{-1}$) that may be estimated from the energy balance equation, simplified here^b as

$$\frac{F_{\odot}}{r_H^2}(1 - A) = \chi (\varepsilon \sigma T^4 + f_s(T)H(T)). \quad (3)$$

$F_{\odot} = 1,360 \text{ W m}^{-2}$ is the solar constant, r_H is the heliocentric distance expressed in AU, A and ε are the Bond albedo and emissivity of the ice surface (we assume $A = 0$ and $\varepsilon = 1$), T is the temperature and $H(T)$ is the latent heat of sublimation of the ice. The term on the left is the power per unit area absorbed from the Sun. The first term on the right is the power per unit area radiated by the ice into space and the second term on the right is the power per unit area used to break intermolecular bonds in the ice and so to drive sublimation. Parameter $1 \leq \chi \leq 4$ is a dimensionless number used to represent the distribution of the absorbed power over the nucleus surface. $\chi = 4$ corresponds to the case where solar power is absorbed over πr_n^2 but radiated from $4\pi r_n^2$, as would be the case for an isothermal sphere of radius r_n . $\chi = 1$ corresponds to the case where the absorbing and radiating (and sublimating) areas are the same, as would be the case for a plane surface oriented normal to the Sun-comet line (e.g., as at midday).

Equation (3) contains too many variables to be solved by itself, so must be considered simultaneously with a relation for $f_s(T)$, provided by the Clausius-Clapeyron equation from thermodynamics. Solutions for water, carbon dioxide and carbon monoxide ices are shown in Fig. 2. In the limit as $r_H \rightarrow 0$, the sublimation term in Eq. (3) dominates over the radiation term (because sublimation is exponentially dependent on T , whereas radiation is proportional only to T^4) and we may write

$$f_s \sim \frac{F_{\odot}(1 - A)}{\chi H(T)r_H^2}. \quad (4)$$

The $f_s \propto r_H^{-2}$ dependence in Eq. (4) is evident in all the curves in Fig. 2 when $r_H \lesssim 2 \text{ AU}$, precisely where most comet observations are

^bA more detailed calculation would balance the energy at each element of the surface and include a term for thermal conduction into the interior. Nothing would be learned for our present purposes by pursuing this more detailed calculation.

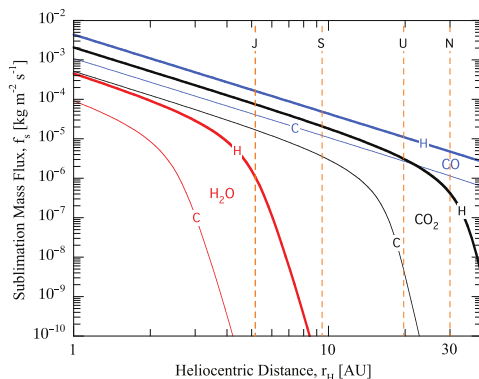


Fig. 2. Equilibrium sublimation rates of three common ices as a function of heliocentric distance. We show water ice, carbon dioxide ice and carbon monoxide ice as labeled. Each volatile is shown as two curves labeled “H” and “C”, for hot and cold temperature limits as described in the text. The orbits of the giant planets are marked by dashed vertical lines.

taken. For water ice, the least volatile of the plotted ices, f_s shows a sharp downturn at larger r_H , defining the region where comets are most active. For CO ice, the downturn lies beyond the orbit of Neptune, and $f_s \propto r_H^{-2}$ applies across the entire planetary system. CO₂ ice is intermediate in volatility and behavior.

Armed with the solutions to Eq. (3) we can make a crude guess as to the lifetimes of comets to the loss of volatiles. For a spherical nucleus of radius r_n sublimating from an area $4\pi r_n^2 f_A$, the time over which the entire nucleus would sublimate away is

$$\tau_{\text{sub}} \sim \frac{\rho_n r_n}{3 f_s f_A}, \quad (5)$$

where $0 \leq f_A \leq 1$ is the fraction of the nucleus surface which is losing mass. Measurements suggest $f_A \sim 10^{-2}$ (e.g., as in Ref. 6), but a wide range of values exists in the active comets. From Fig. 2, water has $f_s \sim 3 \times 10^{-5} \text{ kg m}^{-2} \text{ s}^{-1}$ at $r_H = 3 \text{ AU}$ and, with $\rho_n = 500 \text{ kg m}^{-3}$, a 1 km radius nucleus would last $\tau_{\text{sub}} \sim 10^{12} \text{ s}$ (about 10^5 years). This is already longer than τ , and is a strong lower limit to the true

sublimation lifetime because comet orbits are eccentric, and nuclei spend most of the time at larger r_H where f_s is greatly reduced. Moreover, comets grow refractory “mantles”, consisting of blocks too large to be lifted by gas drag, causing a gradual throttling of the outgassing. Volatile depletion by outgassing in comets is evidently rather slow.

It turns out that outgassing can destroy comets in a less direct but much more devastating way. Outgassing, strongest on the hot dayside of the nucleus, exerts a net reaction force. The resulting non-gravitational acceleration is readily measurable on the orbital timescale (and, indeed, was the observational foundation of Whipple’s model of the cometary nucleus⁷). Significantly, the reaction force vector does not in general pass through the center of mass of the nucleus, creating a torque which can change the spin, both in magnitude and direction. A sustained torque can drive the spin to the limit, where centripetal forces can no longer be met by gravitational forces to the center of a rotating body. For a sphere, this critical period is given by $P_c = (3\pi/(G\rho_n))^{1/2}$ and, for $\rho_n = 500 \text{ kg m}^{-3}$, is $P_c \sim 4.7$ hours. A nucleus spinning with a shorter period is liable to break up unless held together by frictional or cohesive forces.

The spin-up time is $\tau \sim L/T$, where L is the angular momentum and T is the torque. Neglecting considerations of shape, for the moment, we note that $L \propto M_n r_n^2$ which, because $M_n \propto r_n^3$, gives $L \propto r_n^5$. The torque is proportional to the mass loss rate from the nucleus multiplied by the moment arm of the torque. Since mass loss rate is $\propto f_s f_A r_n^2$, and the moment arm $\propto r_n$, we see $T \propto r_n^5 / r_n^3 \propto r_n^2$. Therefore, from dimensional considerations alone, we expect $\tau \propto r_n^2$. The more detailed relation for a spherical nucleus is⁵

$$\tau = \left(\frac{4\pi}{15}\right) \left(\frac{\rho_n r_n^2}{k_T(r_n) f_A(r_n) V_{th} P}\right) \left(\frac{1}{\overline{f_s(r_H)}}\right), \quad (6)$$

where $\overline{f_s(r_H)}$ is the mass loss rate per unit area averaged around the cometary orbit, V_{th} is the speed of the outflowing gas, P is the instantaneous nucleus rotation period and $0 \leq k_T \leq 1$ is the “dimensionless moment arm” for the torque. Physically, k_T is the fraction of the outflow momentum that exerts a torque on the nucleus. Limiting values are $k_T = 0$ for isotropic outflow and $k_T = 1$ for purely collimated, tangential outflow.

Amazingly, we now possess good enough data to detect rotational period changes in comets, as shown in Fig. 3. Even more amazingly, the data are consistent with $\tau \propto r_n^2$ as expected from dimensional considerations; large nuclei are relatively unaffected by outgassing torques while small ones can be strongly affected. The empirical relation is

$$\tau \sim 100 \left(\frac{r_n}{1 \text{ km}} \right)^2, \quad (7)$$

with τ in years and r_n in km. This relation is strictly applicable only to comets with orbits like those of the objects in Fig. 3, i.e., SPCs with perihelia in the range 1 to 2 AU, but it is sufficient to show that the lifetime to spin-up can be very short. The empirical median dimensionless moment arm is $k_T = 0.007$. The very short sub-kilometer cometary nucleus spin-up times explain why sub-kilometer nuclei are rare.

Larger comets cannot be so easily destroyed by rotational instabilities and linger until they are removed by the loss of volatiles or scattering by a planet (Fig. 4). The end state of volatile loss should

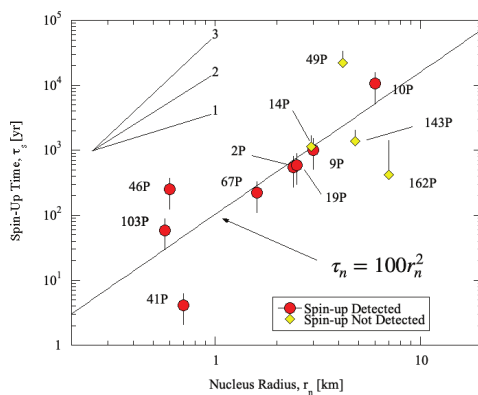


Fig. 3. Empirical spin-up times for comets as a function of radius. Filled circles indicate comets where spin changes are detected, while diamonds indicate that only upper limits to spin changes are available. Adapted from Ref. 5.

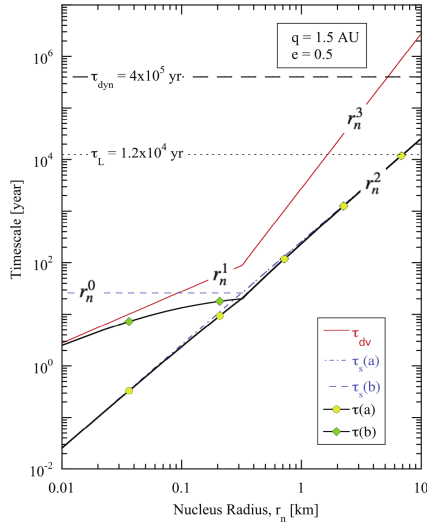


Fig. 4. Model lifetimes for comets as a function of radius including the effects of volatile loss and spin-up instability. From Ref. 5.

be a dead comet, with an asteroid-like physical appearance but moving in the orbit of a comet. Such objects are known, but they are not as abundant as estimates of the relevant timescales would suggest, presenting a puzzle for future investigation.

3. Asteroids

3.1. Spectral and compositional gradients

Asteroids in the main belt exhibit a wide range of surface properties. They are radially segregated from the inner edge (~ 2 AU), where highly metamorphosed S-types (S for “stony”, having optically reddened surfaces and albedos ~ 0.15) dominate, to the outer edge near

3 AU, where less heated C-types (C for “carbonaceous”, colorimetrically more neutral, with low albedos ~ 0.05) are more common. The gradient is too strong to be explained by the decrease of radiation equilibrium temperature from 2 to 3 AU. An early explanation for this compositional stratification invoked the varying influence of induction heating in an early, highly magnetic phase of the Sun. It is more widely accepted that strong heating of asteroids occurred through the decay of ^{26}Al (half-life 0.7 Myr, see Ref. 9). One way to cook inner belt asteroids more thoroughly than outer belt objects is to delay the accumulation of the latter relative to the former. In this picture, the radial gradient results from early-forming inner-belt objects being more strongly heated by ^{26}Al trapped in minerals relative to late-forming outer belt asteroids where the ^{26}Al has already decayed away. This explanation is not wholly unreasonable, because longer formation times automatically occur at larger distances in a protoplanetary disk whose density (and hence collision rate) falls with radius. Perhaps the asteroid belt compositional gradient simply reflects a formation time gradient. Various hydrated minerals in meteorites from the outer belt indicate chemical alteration reactions in liquid water, presumably from radioactively melted ice. Recent discovery of the so-called “main-belt comets” (see what follows) suggest that water ice has survived in some objects, preferentially those in the outer belt at $\gtrsim 3$ AU. The contrast with inner-belt S-types, some of which have been heated to 800°C , is extreme.

Isotopic measurements of meteorites provide surprising evidence of a more complicated story. An isotopic dichotomy (Fig. 5), as opposed to a continuous spread, suggests that the asteroids formed in two spatially distinct reservoirs experiencing different thermal histories.⁸ The two groups also appear to be different in age, with the carbonaceous chondrites forming 1–2 Myr after the non-carbonaceous chondrites. The terrestrial planets (Earth, Moon and Mars — we have no known samples of Mercury or Venus) isotopically resemble the LL chondrites (S-type residents of the inner asteroid belt) and the HEDs (ejecta from large asteroid Vesta, semi-major axis 2.4 AU). Carbonaceous chondrites are more associated with the outer belt, where the C-types and main-belt comets reside. One idea is that the two reservoirs formed separately on either side of Jupiter’s orbit. Runaway growth then scattered and implanted material from beyond its orbit into the outer asteroid belt. However, it is difficult to see

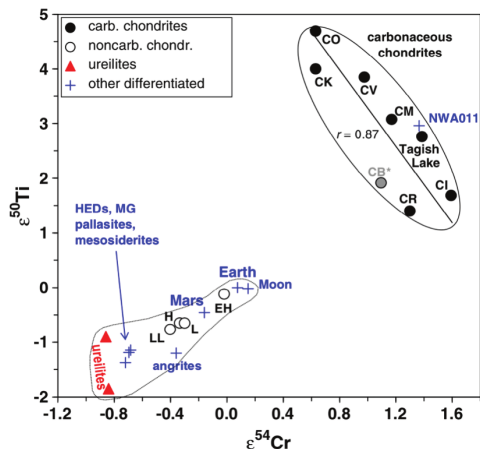


Fig. 5. Fractional isotopic concentrations in titanium and chromium show a bimodal distribution. Adapted from Ref. 8.

how any implantation mechanism would lead to fine compositional distinctions across the minuscule ~ 1 AU width of the asteroid belt.

3.2. Active asteroids

Figure 6 shows the distribution of asteroids and comets in the semi-major axis vs. eccentricity plane. Shown in red are a set of newly-identified objects known as “active asteroids” because, while their orbits are asteroidal, their physical appearances resemble comets because of the intermittent ejection of dust. In addition to being a surprising and newly-identified population, the active asteroids are scientifically interesting because they result from an amazing suite of different physical processes capable of ejecting dust.

3.2.1. Impact

Collisions underlie some of the active asteroids. Collision velocities in the asteroid belt, $\Delta V \sim 5 \text{ km s}^{-1}$, are a significant fraction of the

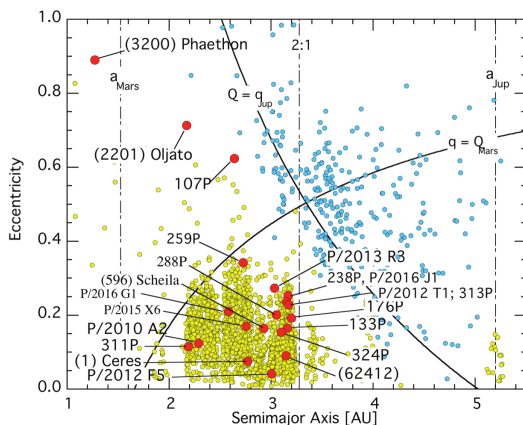


Fig. 6. Distribution of comets (blue dots) and asteroids (yellow dots) in the semi-major axis vs. eccentricity plane. Red circles show active asteroids, defined as objects with the orbits of asteroids but physical appearances like comets, resulting from mass-loss. Dashed, vertical lines mark the orbits of Mars and Jupiter, and the location of the 2:1 mean motion resonance with Jupiter. The diagonal arcs show the loci of points having aphelion equal to the perihelion distance of Jupiter and the perihelia equal to the aphelion of Mars. Objects above the arcs cross either the orbits of Mars or Jupiter and are, consequently, short-lived.

orbital velocity, $V_K \sim 20 \text{ km s}^{-1}$, reflecting the non-negligible average inclinations ($\sim 15^\circ$) and eccentricities (~ 0.15) of the asteroids. The kinetic energy of impact is more than enough to vaporize the projectile provided speed $V > (2H)^{1/2}$, where H is the latent heat of vaporization. For example, with representative $H = 2 \times 10^6 \text{ J kg}^{-1}$, we see that collisions with $V > 2 \text{ km s}^{-1}$ should involve substantial vaporization. With ΔV also being comparable to or greater than the speed of sound in rocks, the effect of impact is to shock process, vaporize and pulverize the asteroids, releasing a cascade of debris into the interplanetary space. 354P/(2010 A2)¹⁰ is the first identified impact-produced active asteroid. Although collisions have long been known as the main agent shaping the asteroid size distribution (an inverse radius³ power law produced by shattering into smaller

and smaller pieces), the interval between major collisions per object is enormous ($\gtrsim 10^7$ years in the case of 354P¹¹). It is startling to be able to detect such rare events in nearly real-time, and a reminder of the vast numbers of small asteroids and collisions ongoing in the belt.

3.2.2. Rotational instability

Whereas rotational instability in comets results from outgassing torques, in asteroids it is the “YORP” torque that plays the same role. The YORP torque results from the anisotropic radiation of infrared photons from an irregularly shaped, anisothermal body. It is $\sim 10^5$ weaker than outgassing torques and so takes $\sim 10^5$ times longer to act, on objects of a given size. Despite this, the time needed to approach rotational instability (a few to 10 Myr for a 1-km diameter asteroid at 3 AU) is still very short compared to the collisional lifetimes of these bodies. Several active asteroids are products of rotational instability, including 313P. Recent discoveries of main-belt asteroids which eject mass or even spontaneously fragment show that rotational break-up occurs and perhaps dominates asteroid destruction at small scales.

3.2.3. Thermal destruction

Differential thermal expansion creates stresses large enough to crack rocks. The fractional expansion resulting from temperature change ΔT is $\alpha \Delta T$, where expansivity $\alpha \sim 10^{-5} \text{ K}^{-1}$ is typical. The resulting strain is $S \sim Y \alpha \Delta T$, where Y is Young’s modulus and a multiplier of order unity called Poisson’s Ratio is ignored. Rock values of $Y \sim 10^{10} \text{ N m}^{-2}$ to 10^{11} N m^{-2} are so large that huge stresses, $S \sim 10^5 \Delta T$ to $10^6 \Delta T \text{ (N m}^{-2}\text{)}$, can be developed. These stresses can overwhelm the strengths of the rocks, which are especially weak in tension (e.g., the tensile strength of basalt is only $\sim 4 \times 10^6 \text{ N m}^{-2}$). Thermal fracture operates most effectively on length scales comparable to the diurnal skin depth (i.e., where the temperature gradient is steepest) which, for a material of diffusivity κ on a body rotating with period P is $\ell \sim (\kappa P)^{1/2}$. For solid rock, $\kappa \sim 10^{-6} \text{ m}^2 \text{ s}^{-1}$ and a typical rotation period $P = 5 \text{ h}$, $\ell \sim 15 \text{ cm}$. Small rocks are thus strongly affected while larger boulders on asteroids are often surrounded by skirts of

debris produced by thermal cracking of their surfaces owing to the day–night temperature cycle.

Asteroid (3200) Phaethon, which is the source of the Geminid meteoroid stream, is the first object showing convincing evidence of particle ejection caused by thermal fracture. Phaethon has a very asteroidal Tisserand parameter (Eq. (1)) $T_J = 4.5$ and an orbit that is distinct from the comets in being completely decoupled from Jupiter. The perihelion is a remarkably small $q = 0.14$ AU, at which distance the sub-solar surface temperature approaches 1,000 K, while the local midnight temperature might be only 400 K, giving $\Delta T \sim 600$ K every rotation period of 3.5 h). Phaethon emits dust particles and sports a comet-like tail when near perihelion.

The distribution of perihelion distances of the asteroids shows a depletion (lower curve in Fig. 7) relative to the expectation based on dynamical models (upper curve). The depletion begins at surprisingly large perihelion distances $q \sim 0.5$ AU (peak sub-solar temperature ~ 500 K) and affects C-types more than S-types. While currently lacking a widely-held explanation, this near-Sun depletion is very plausibly a result of thermal disintegration.

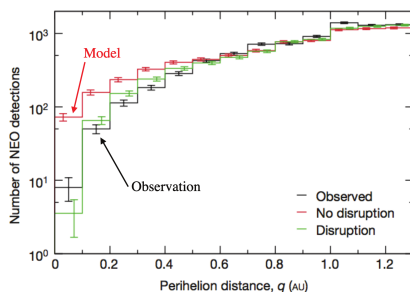


Fig. 7. Observed and expected numbers of near-Earth objects as a function of the perihelion distance. The two curves diverge at perihelion distances $\lesssim 0.5$ AU, conceivably indicating an unknown error with the dynamical model but, more likely, a physical destruction of asteroids with small perihelia. Adapted from Ref. 13.

3.2.4. Sublimation

Some active asteroids, including the prototype object 133P/Elst-Pizarro,³ appear to be driven by sublimation of embedded but near-surface ice; these are the main belt comets (MBCs). The evidence for this remains indirect, since sublimated gas production rates are orders of magnitude too small to be detected by existing spectroscopic techniques (JWST might change this), but nevertheless the distinctive properties of these objects are compelling. Most distinctive is the reappearance of dust activity near perihelion in different orbits, which establishes that the activity is thermally driven just as in “normal” comets from the Kuiper belt. The big difference between the normal comets and the active asteroids is that the latter have relatively circular orbits at 3 AU, and surface temperatures that are far too high (the subsolar temperature at 3 AU is ~ 225 K) for the ice to survive for even a tiny fraction of their $\sim$ billion year collisional destruction ages. For example, Fig. 2 shows that water ice exposed at the sub-solar point on an asteroid at 3 AU sublimates at $f_s \sim 3 \times 10^{-5}$ kg m $^{-2}$ s $^{-1}$. If the density is $\rho = 10^3$ kg m $^{-3}$, a patch of ice would shrink at the rate $dr/dt = f_s/\rho = 3 \times 10^{-8}$ m s $^{-1}$. This rate sounds small (coincidentally, it is about the same as the peak rate of a child), but a 1-km radius asteroid would, in the absence of intervention, last only $\sim 10^3$ years given this dr/dt . We surmise that the ice, if present, must be stabilized against sublimation losses. A simple layer of dirt a few meters thick suffices to suppress the temperature at the buried ice surface and to inhibit the flow of sublimated gases to the surface.

The scenario, then, is that MBCs are ice-containing asteroids in which the ice remains buried below the physical surface under a fragmented layer of refractory debris. Disturbances of this layer, by small impactors or perhaps by surface avalanches induced by rotational instability, occasionally expose the ice to the heat of the Sun, causing sublimation and the production of a temporary coma. After some time, the active surface again becomes clogged by refractory material and the ice recedes beneath the physical surface, stopping further sublimation. A key parameter in this conjectural model is the “duty cycle”, the fraction of the total time spent in the active state. Measurements show that the duty cycle must be $< 10^{-4}$ and is probably $< 10^{-5}$, allowing the ice to survive for long timescales. Crucially, these small duty cycles also mean that each MBC is representative of 10^4 or

10^5 inactive but otherwise similar inactive ice-containing asteroids. Given this, it is possible that the entire outer asteroid belt consists of ice-containing objects.

Where did the MBCs come from? Implantation from the modern-day Oort cloud and Kuiper belt is possible but energetically very unlikely because too much energy dissipation is required. Implantation from a middle or outer solar system source seems to require a different arrangement of planets giving resonant scattering and low probability trapping, as has been suggested for the Jupiter Trojans. It is equally possible that the MBCs formed in-place. Even though the asteroid belt is currently inside the effective snow-line of the solar system, we know that the snow-line moved in response to time-dependent changes in the dominant heating sources in the Sun's protoplanetary disk. Perhaps these asteroids formed, as with the other comets, outside the snow line and then managed to survive for most of the age of the solar system after the disk cleared and the asteroid radiation equilibrium temperatures equilibrated to their modern values.

Why does this matter? One reason is that the outer belt is the most likely source region for the terrestrial planet volatiles, which appear to have been delivered at least in part by late stage collisions occurring after the magma oceans had cooled. If this is correct, the MBCs are disk remnants of the ocean precursors.

4. Kuiper Belt

Figure 8 shows the distribution of outer solar system bodies in semi-major axis vs. eccentricity space. This space is clearly not uniformly occupied, with obvious clumping into five major groups. In the figure, "C" marks the classical KBOs, so labeled because their modest eccentricities (and inclinations) most closely resemble ideas about the Kuiper belt viewed as an accretion disk, prior to its discovery. Still, though, even the classical objects occupy a distribution of e and i broader than expected of an accretion disk, indicating that their orbits have been dynamically excited. Close examination of the " i " distribution reveals two types of classical objects, distinguished by inclinations less than and greater than a few degrees ($i < 5^\circ$ is commonly used, but there is no sharp boundary). The former, so-called

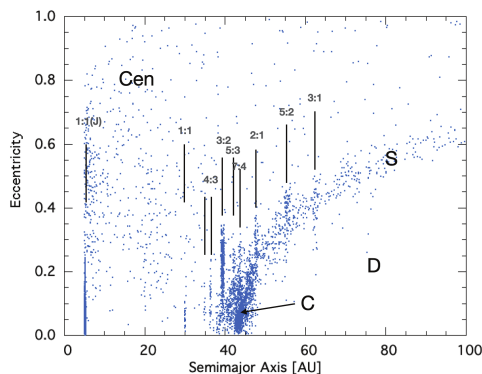


Fig. 8. Outer solar system objects in the semi-major axis vs. eccentricity plane. *C* = classical KBOs, *S* = scattered disk objects, *D* = detached objects, *Cen* = Centaurs. The locations of numerous mean-motion resonances with Neptune are marked. Pluto is in the 3:2 at $e = 0.24$.

cold-classical KBOs, are the reddest objects with the largest fraction of known binaries, and are believed to be more primitive than the dynamically excited “hot classicals” with their reduced number of ultrared surfaces and binaries. The classical KBO belt has an outer edge near the 2:1 mean motion resonance at 47 AU that is not explained.

The “S” or scattered objects have perihelia in the 30–40 AU range, where perturbations from Neptune are sufficient to excite the orbits on 10^9 year timescales. Successive perihelion interactions with Neptune pump the eccentricities and raise the perihelia leading, eventually, to these objects being detached by external perturbations from passing stars. The long-term chaotic orbits of the scattered KBOs provide a source of Neptune-crossing bodies¹³ which, in turn feed a flow of objects toward the terrestrial planet region. Scattered KBOs with perihelia between Jupiter and Neptune are known as Centaurs, labeled “Cen” in the figure. Centaurs that cross Jupiter’s orbit are relabeled as comets based on their rising levels of outgassing activity. Population “D” in the figure is a set of KBOs whose perihelia are too far from Neptune for this planet to have substantially affected their

orbits. The “D” stands for detached, and the origin of the detached objects is currently a mystery. Ancient perturbations from a passing star have been suggested to have raised their perihelia from (presumably) the smaller orbits in which they formed, as have perturbations from unseen sub-planet and planet-sized bodies. The most remarkable feature of Fig. 8 is the abundance of vertically stacked KBOs. These are the resonant objects, with the Plutinos being those Pluto-like objects in the 3:2 mean motion resonance at 39 AU. The high abundance of resonant KBOs (relative, for example, to the number expected if objects were randomly sprinkled onto the $a-e$ plane) points to resonance sweeping by outward migration of Neptune.

4.1. *Planetary migration and late-heavy bombardment*

The discovery that large numbers of KBOs are trapped in mean-motion resonance with Neptune has changed the way in which we view the evolution of the Solar System. This is because the most plausible resonance trapping mechanism requires the outward radial migration of Neptune and, by implication, the radial migration of all the planets. In turn, radial migration upsets the classical “clockwork” view of the Solar System in which the orbits of planets are fixed and strictly periodic. Radial migration allows the distances and interactions between planets to be highly time variable, with the possibility that some major planets themselves might have once crossed mean-motion resonances causing dramatic disturbances of solar system architecture. This is the idea behind the Nice model, which was proposed to explain the “late-heavy bombardment” (LHB) recorded in the crater age distribution of the Moon.¹⁴ These authors arranged for the torques from a $30 M_{\oplus}$ Kuiper belt (the current mass is near $0.1 M_{\oplus}$) to drag Neptune outwards, causing Jupiter and Saturn to resonantly interact about 700 Myr after the formation of the Solar System, this time coinciding with the estimated date of the LHB. Destabilization of the Kuiper belt and other objects in the Solar System would shower the Moon and other terrestrial bodies with impactors, leaving the LHB spike. The disturbance of the primordial Kuiper belt is also supposed to have populated other dynamical niches in the Solar System, including the Trojans of Jupiter, the irregular satellites of the planets, and perhaps even the asteroid belt.

However, it was known even in the 1970s that the LHB could be explained another way, without recourse to a special impacting spike at 3.8 Gyr (readable summaries of the issues are given in Refs. 15 and 16). Essentially, the concordance of the ages of several large basins, as sampled by the rocks from the Apollo landing sites, could reflect Moon-wide blanketing of the surface by ejecta from the latest large impact. This would mean that Apollo missions to widely separated sites on the Moon might really be sampling material from a single large event at 3.8 Gyr, instead of representing a local spike in the impact rate at that time. New evidence from lunar zircons, which are ancient, highly robust silicate crystals in which the chemistry records the history of heating and shock processing, shows that this alternative explanation is the correct one.¹⁷ There is no spike in the age distribution of zircons as would be expected if a late-heavy bombardment had occurred. Since this finding, the parameters assumed in the Nice model have been adjusted to induce resonance crossing closer in time to the origin of the solar system, where a spike of impactors would be lost in the naturally heavy impact rates prevailing as the solar system cleared.¹⁸ This is a classic example of how a multi-parameter model can fit a wide range of constraints, even ones that turn out to be invalid.

4.2. *Unseen planet*

Trujillo *et al.*¹⁹ noticed that the orbits of large semi-major axis KBOs appear to be non-randomly distributed and suggested that this could result from perturbations by an unseen planet in the outer solar system. The distribution of the argument of perihelion, ω , for instance, is shown in Fig. 9. The large semi-major axis objects appear to avoid $60^\circ \lesssim \omega \lesssim 240^\circ$, as emphasized in the third panel of the figure. Whether or not this effect indicates the presence of an unseen planet remains unclear. First of all, not all directions in the sky can be equally well observed, and accounting for the role of observational bias in producing the non-random distribution is difficult and disputed. KBOs in parts of the sky overlapping the Milky Way, for example, cannot be efficiently detected, while other less obvious biases may be important (c.f. Ref. 20). Second, even if the non-random distribution is real, it might have a cause unrelated to

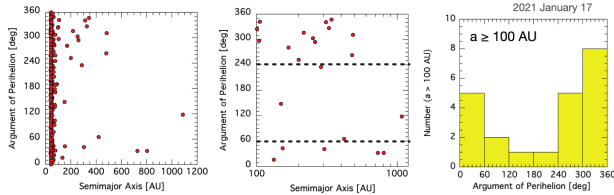


Fig. 9. Evidence for non-uniform distribution of large semi-major axis KBOs. (left) Semi-major axis vs. argument of perihelion (middle), same, but zoomed to exclude objects <100 AU (right) histogram of the objects from the middle panel. Data from JPL Small Body Database.

an unseen planet. The question of whether or not an unseen planet exists will persist until such an object is directly observed.

4.3. Giant planet Trojans

Only Jupiter and Neptune are known to have Trojans, separated from their parent planets in longitude by $\pm 60^\circ$ and librating around the L4 and L5 Lagrangian points. There are two basic ideas for the origin of the Trojans. First, they might be planetesimals that accreted in heliocentric orbits in the vicinity of the giant planets and which were subsequently captured, most likely by 3-body interactions in the vicinity of the Lagrangian points. In this case, the compositions of the Trojans would preserve local material, most of which was consumed by the planets during their growth (to make their heavy-element cores). Alternatively, the Trojans might have been captured from an outer Solar System source during the instability formerly employed to explain the LHC but now pushed back to a much earlier time. The material of the Trojans would, in this case, preserve material formed at much larger distances and lower temperatures, perhaps in the KB itself.

At the present time, there is not much direct evidence to support one conjecture over the other. Local capture into the Trojan swarms is energetically highly preferred, because bodies in Jupiter-like orbits have small approach velocities and require little energy dissipation in order to be captured (the escape velocity from the Trojan swarms is

only $\sim 100 \text{ m s}^{-1}$ and relative velocities must be damped below this value for capture to occur). Bodies scattered from the KB, on the other hand, are falling in to Jupiter's distance with a large fraction of the local free fall velocity ($\sim 20 \text{ km s}^{-1}$) and so have a very low (10^{-5} to 10^{-6}) probability of capture.

One property that is consistent with capture from a distant source is the Trojan inclination distribution, which is very broad, with a half-width at half maximum of 15° . This broad distribution is preserved from the Kuiper belt source in distant capture models²¹ whereas, otherwise, some mechanism of inclination pumping (e.g., gravitational scattering by large embedded embryos, which no longer survive) must be invoked to fit the observations.²² It has been claimed that the size distributions of the large Trojans and the KBOs are identical, but this claim is difficult to sustain because there is almost no overlap between the distributions measured at the two populations as a result of observational bias (e.g., a given object at 40 AU is $(40/5)^4 \sim 4,000$ times fainter than at 5 AU). The color distribution of the Jovian Trojans is unlike that of the Kuiper belt source region (typified by the scattered disk objects), requiring some process of surface modification if the Trojans are escaped KBOs. Perversely, the same modification appears to have affected Neptune's Trojans, which are colorimetrically like the Jupiter Trojans and distinct from the supposed Kuiper belt source.^{23,24} We hope that NASA's Lucy mission to the Trojans will throw new light on their origin.

5. Q&A

Max Mahlke: Do you think it still makes sense to divide asteroids and comets into separate populations and, if so, how should it be done?

David Jewitt: There is practical value in having a distinction, but there are certainly objects at all positions along the way between the extremes. If you are looking for an absolute definition of asteroid vs. comet, then there isn't one. There is no single good definition, and different definitions tend to be used by different people. In the end, the definition is not that crucial, in my view, any more than whether you describe Australia as continent or an island.

Julia Maia: Do the active asteroids have a particular color and, when a body becomes active, does the color change? You might expect that, for example, if the surface is space-weathered, but the interior is not.

David Jewitt: In general, the active asteroids that probably contain ice tend to be neutral colors, like C-type asteroids. Asteroid Gault, which I think is losing mass because it is near rotational instability, was reported to have changed its color. The change is subtle, though, and I wonder if this is some problem of measurement. But there really isn't much relevant data for other objects.

Denis Shulyak: Does the flux of comet impactors scale with the mass of the disk, how does it depend on the distribution of the planets, does Jupiter protect Earth or focus comets to hit us?

David Jewitt: We can't give a general answer. But, sure, if you took the solar system as it is and increased the number of Kuiper belt objects by a factor of 100, you'd increase the impact rate by the same factor. But comparing different planetary system geometries would change everything and I don't know how to address that case without somebody doing dynamical simulations. We do know, in our case, that the comets are not a big threat compared to the asteroids. The impact threat from comets is about 1% of the threat from asteroids. So, for us, the comet impact threat (see the 2021 movie *Don't Look Up*) doesn't matter much.

Natalia Rossignoli: Did the New Horizons flyby of Arrokoth reveal the nature of the red material or show that the ice is amorphous?

David Jewitt: Unfortunately, no. The big problem in the outer solar system is that the spectra of most bodies are featureless (possibly because they have been devolatilized or carbonized by cosmic rays). NH was good at showing the body shape of Arrokoth, and it did detect methanol. But methanol was already known in a few other KBOs over the last couple of decades, so it was not a big surprise. The spacecraft provided no constraint on the crystalline state of the water, though, because water was not even detected.

Patricio Zain: Are there estimates of how much mass is lost when an asteroid breaks up through a rotational instability?

David Jewitt: From observations, yes, but they're very uncertain, because we cannot see mass, we only see material through its light scattering cross-section. That means that you can always hide mass in big particles that present relatively small cross-sections, so the observational masses are under-estimates. This is a fundamental problem in the solar system, as it is in studying the debris disks and accretion disks of stars. It's hard to measure the mass. But for example, I estimate that 331P lost $\sim 1\%$ of its mass due to a rotational event. The models, unfortunately, have a large number of assumed parameters, so they don't predict the mass loss. It really goes the other way; the data feed the models, not vice versa.

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